Ex.No.12	Examine Network Address Translation (NAT) using Cisco Packet Tracer
Date:	

AIM

To configure and examine **Network Address Translation (NAT)** using **Cisco Packet Tracer** and verify communication between a private network and a public network.

Requirements

- Cisco Packet Tracer
- 1 Router (Cisco 1941)
- 2 Switches (Cisco 2960)
- 2 PCs
- Copper Straight-Through cables

Network Topology

Private Network Public Network

PC0 -------- Switch0 -------- Router -------- Switch1 -------- PC1

192.168.1.10 NAT Router 203.0.113.10

G0/0 Public PC

192.168.1.1

|

G0/1

203.0.113.1

IP Addressing

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.10	255.255.255.0
Router	G0/0	192.168.1.1	255.255.255.0
Router	G0/1	203.0.113.1	255.255.255.0
PC1	FastEthernet0	203.0.113.10	255.255.255.0

PROCEDURE

Step 1: Create the topology

1. Open **Cisco Packet Tracer**.
2. Place:
    - Two PCs.
    - Two 2960 switches.
    - One 1941 router.
3. Connect:
    - PC0 → Switch0.
    - Switch0 → Router G0/0.
    - Router G0/1 → Switch1.
    - Switch1 → PC1.
4. Use **Copper Straight-Through** cables.

Step 2: Configure PC0

Go to:

PC0 → Desktop → IP Configuration

Enter:

IP Address : 192.168.1.10

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.1.1

Step 3: Configure PC1

Go to:

PC1 → Desktop → IP Configuration

Enter:

IP Address : 203.0.113.10

Subnet Mask : 255.255.255.0

Default Gateway : 203.0.113.1

Step 4: Configure the Router

Open:

Router → CLI

Configure the inside interface:

Router> enable

Router# configure terminal

Router(config)# interface gigabitEthernet 0/0

Router(config-if)# ip address 192.168.1.1 255.255.255.0

Router(config-if)# ip nat inside

Router(config-if)# no shutdown

Router(config-if)# exit

Configure the outside interface:

Router(config)# interface gigabitEthernet 0/1

Router(config-if)# ip address 203.0.113.1 255.255.255.0

Router(config-if)# ip nat outside

Router(config-if)# no shutdown

Router(config-if)# exit

Step 5: Configure NAT

Create an access list to identify the private network:

Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255

Configure **PAT (NAT Overload)**:

Router(config)# ip nat inside source list 1 interface gigabitEthernet 0/1 overload

This allows multiple private IP addresses to share the router's outside IP address.

Step 6: Verify the Router Configuration

Execute:

Router# show ip interface brief

Expected output:

Interface IP-Address Status

GigabitEthernet0/0 192.168.1.1 up

GigabitEthernet0/1 203.0.113.1 up

Step 7: Test Connectivity

From **PC0**, open:

Desktop → Command Prompt

Enter:

ping 192.168.1.1

Expected:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Now test the public-side PC:

ping 203.0.113.10

Expected:

Reply from 203.0.113.10: bytes=32 time<1ms TTL=127

Step 8: Verify NAT Translation

Generate traffic from PC0 by pinging PC1:

PC0> ping 203.0.113.10

Then, on the router, enter:

Router# show ip nat translations

Sample output:

Pro Inside global Inside local Outside local Outside global

icmp 203.0.113.1:1 192.168.1.10:1 203.0.113.10:1 203.0.113.10:1

This shows that the **private IP address 192.168.1.10** has been translated to the router's **public IP address 203.0.113.1**.

Step 9: Verify NAT Statistics

Enter:

Router# show ip nat statistics

Sample output:

Total active translations: 1

Inside interfaces:

GigabitEthernet0/0

Outside interfaces:

GigabitEthernet0/1

Hits: 5

Misses: 0

The exact numbers may vary depending on the number of packets generated.

SAMPLE OUTPUT

PC0 pinging PC1:

PC> ping 203.0.113.10

Pinging 203.0.113.10 with 32 bytes of data:

Reply from 203.0.113.10: bytes=32 time<1ms TTL=127

Reply from 203.0.113.10: bytes=32 time<1ms TTL=127

Reply from 203.0.113.10: bytes=32 time<1ms TTL=127

Reply from 203.0.113.10: bytes=32 time<1ms TTL=127

Ping statistics:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

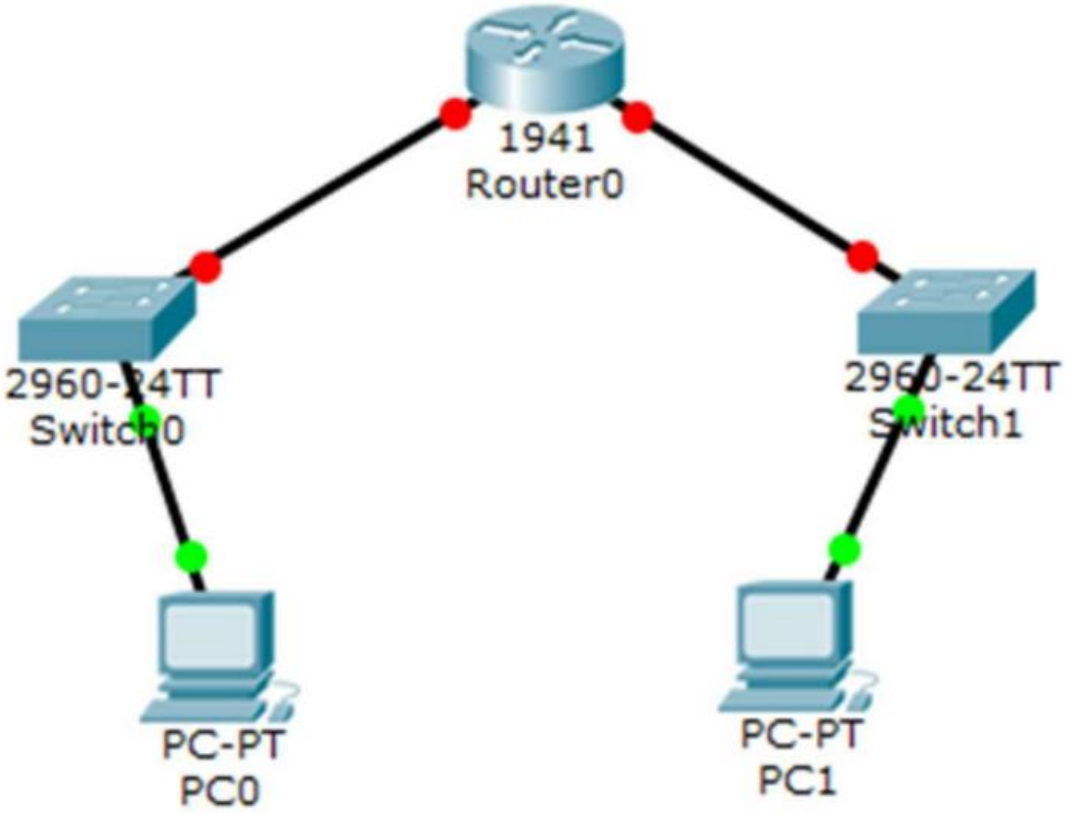
NAT Translation Table:

Router# show ip nat translations

Pro Inside global Inside local Outside local Outside global

icmp 203.0.113.1:1 192.168.1.10:1 203.0.113.10:1 203.0.113.10:1

OUTPUT:



RESULT:

Thus, **Network Address Translation (NAT)** was successfully configured and examined using **Cisco Packet Tracer**. The private IP address **192.168.1.10** was translated into the public IP address **203.0.113.1**, and successful communication with the public network was verified.